METRIC

MIL-PRF-8565/3F
20 October 2021
SUPERSEDING
DOD-B-8565/3E(AS)
26 June 1987

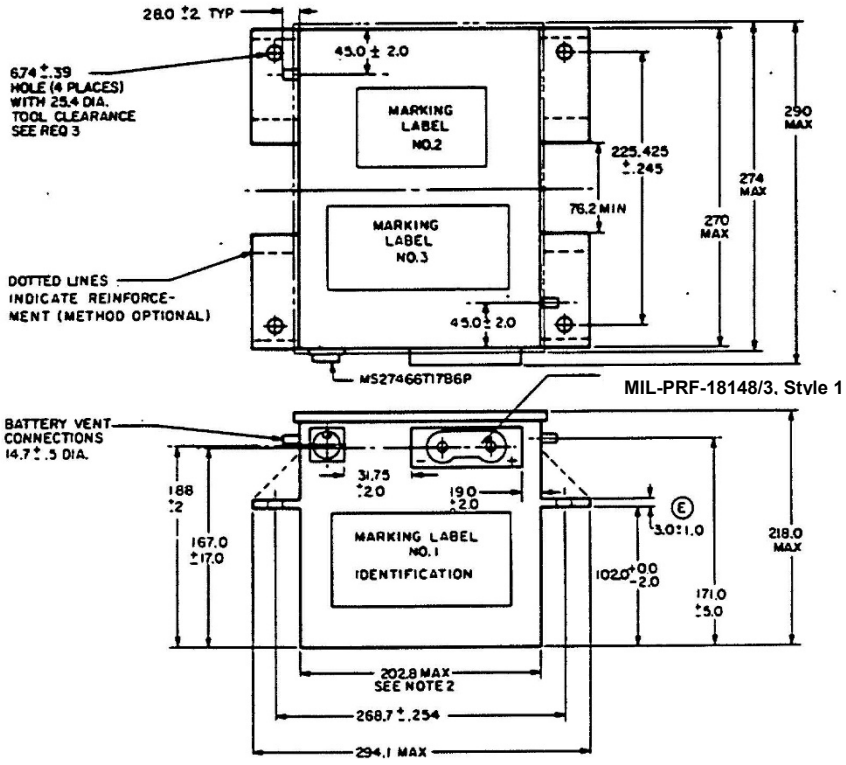
PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM RATE, NO MAINTENANCE, 24-VOLT, 15 AMPERE-HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

D8565/3-2 was inactivated for new design after 26 June 1987.
D8565/3-3 should be used in place of D8565/3-2 for new design.

The complete requirements for acquiring the storage battery described herein shall consist of this document and the latest issue of specification MIL-PRF-8565.

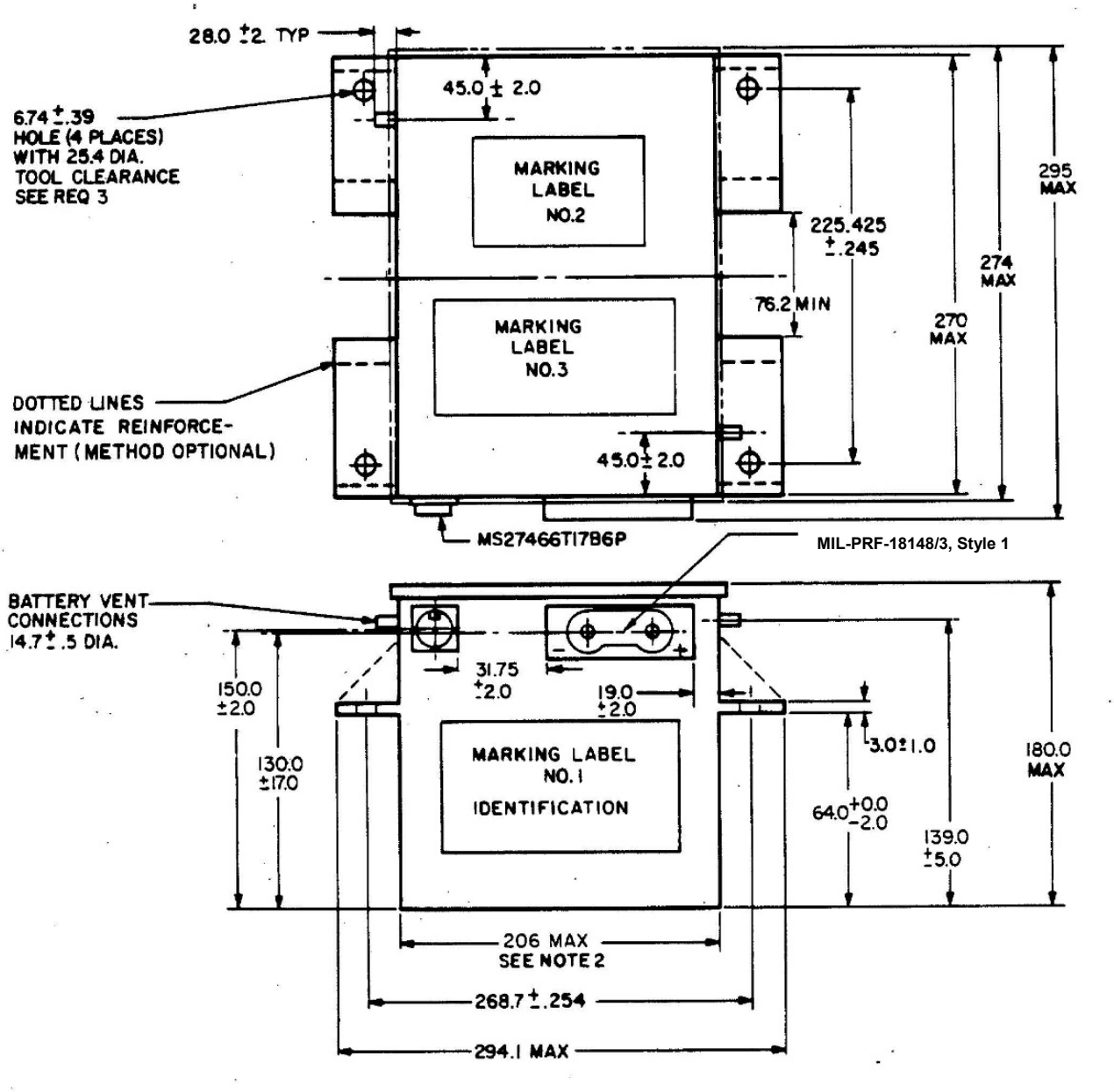


NOTES:

1. All dimensions are millimeters.
2. Allow additional 4 mm width for cover plus 4 mm for mounting bracket across side of container.

FIGURE 1. Battery P/N D8565/3-2.

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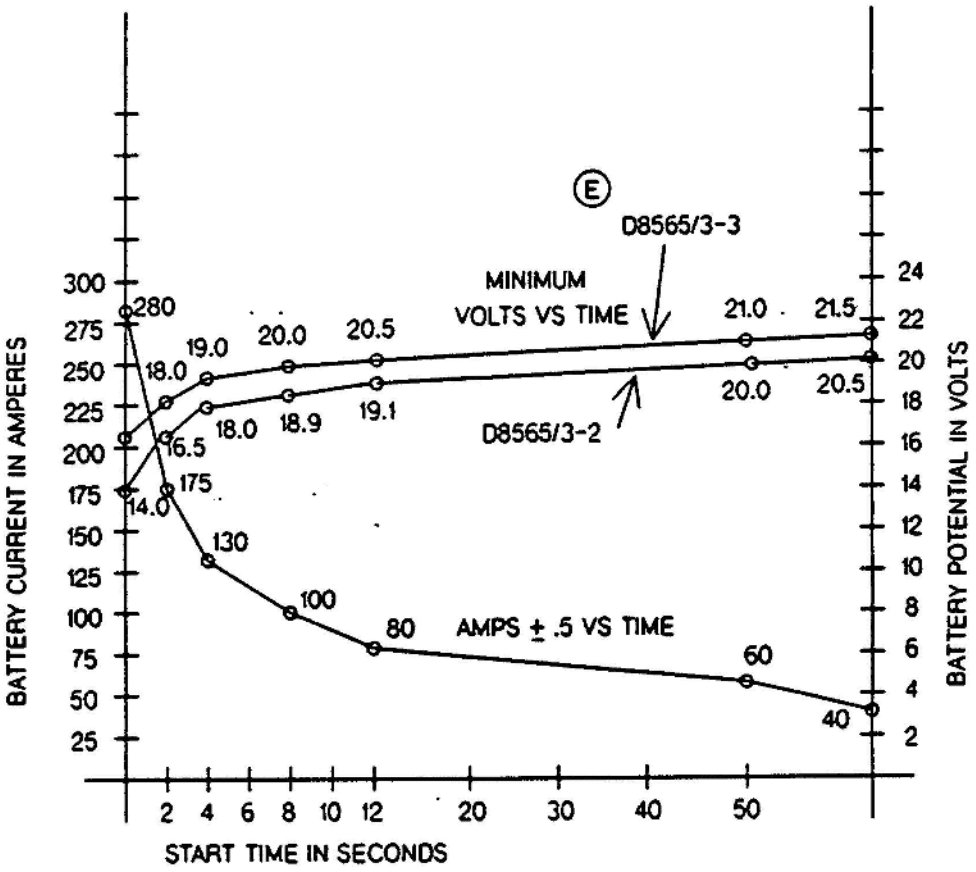


NOTES:

1. All dimensions are millimeters.
2. Allow additional 4 mm width for cover plus 4 mm for mounting bracket across side of container.

FIGURE 2. Battery P/N D8565/3-3.

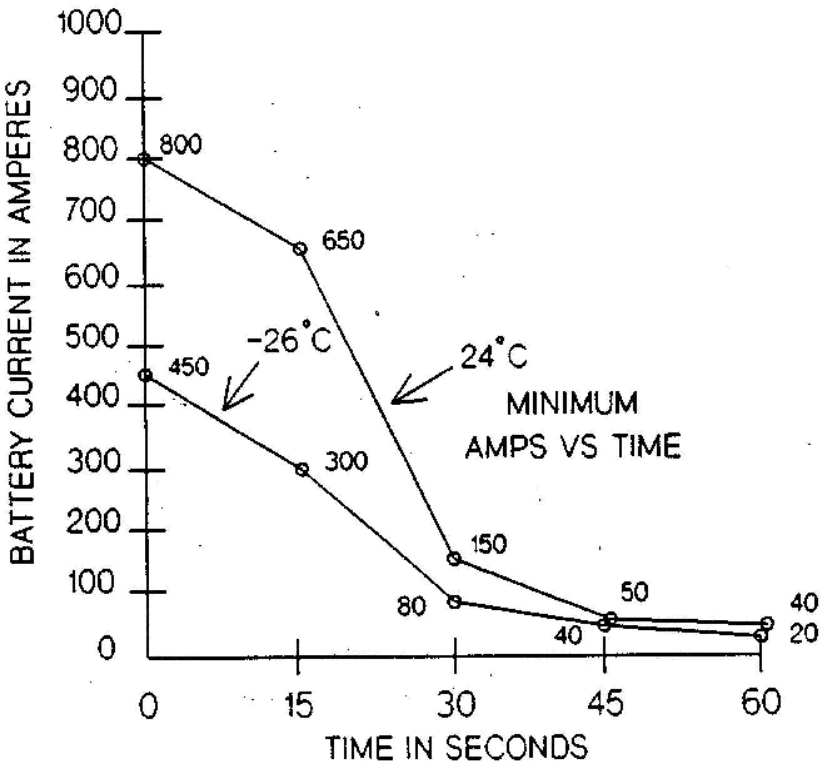
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COORDINATES OF CURVES STRAIGHT LINE BETWEEN COORDINATES			
Current in Amps	Potential in Volts		Start Time in Seconds
	D8565/3-2	D8565/3-3	
280	14.0	16.5	0
175	16.5	18.0	2
130	18.0	19.0	4
100	18.9	20.0	8
80	19.1	20.5	12
60	20.0	21.0	50
40	20.5	21.5	60

FIGURE 3. Start cycle requirements.

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COORDINATES OF CURVES STRAIGHT LINE BETWEEN COORDINATES		
Time Seconds	Current in Amperes at	
	24°C	-26°C
0+	800	450
15	650	300
30	150	80
45	50	40
60	40	20

FIGURE 4. Constant voltage (14.0 V) discharge curves (3.8.7 and 4.5.8).

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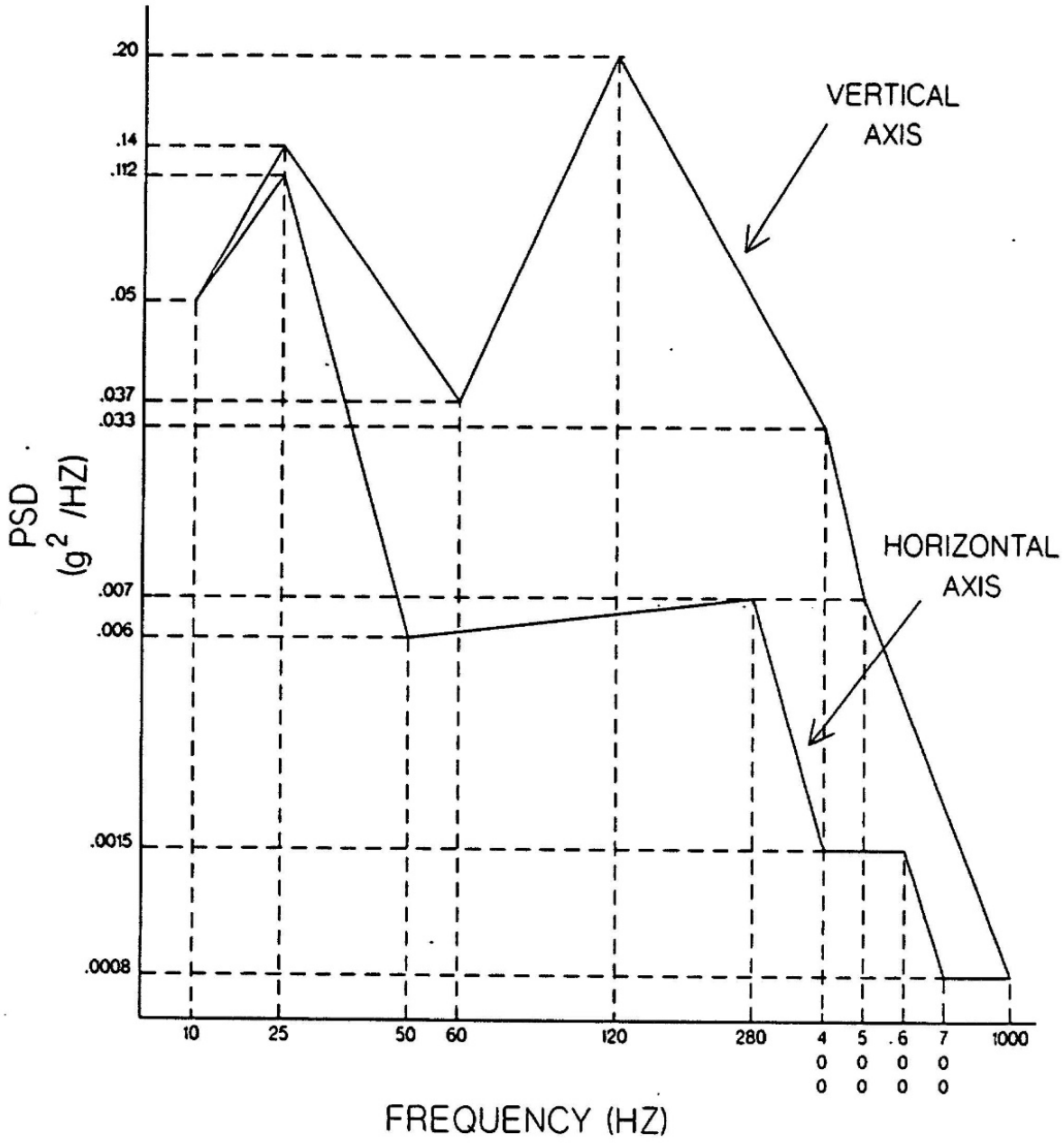


FIGURE 5. Random vibration test levels.

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REQUIREMENTS:

1. The rated capacity of the battery shall be 15 ampere-hours (15.0 Ah/1 hr./21°C/18.0 V at end of life). 1C = 15.0 amperes.

2. The D8656/3-2 battery shall weight 26 kg max. The D8565/3-3 battery shall weight 21.5 kg max.

3. Mounting holes shall have a coating-free surface, 12.7 mm diameter both sides.

4. Aircraft installation. For installation in the aircraft, the battery shall be supported as shown on figure 3 of MIL-PRF-8565. Vibration isolation shall be provided, if necessary, to insure that the vibration level does not exceed the requirements of the 2-year cycling test of 4.5.10, as modified herein. If vibration isolators are used, sufficient sway clearance between the battery and the mounting structure shall be provided. Battery life in the aircraft shall be 2 years.

5. The battery shall conform to military specification MIL-PRF-8565, except as modified by the following:

3.5.5 Receptacles. Add: "The receptacle for the battery's positive and negative connections shall be dimensionally equal to MIL-PRF-18148/3, Style 1."

6. Add the following new paragraph:

3.5.1.1 Add: "An auxiliary receptacle MS27466T17B6P, shall be located as shown on figures 1 and 2.

a. Pin A shall be high side AC (input heater power).

b. Pin B shall be AC ground (heater return).

PIN A 115 VAC

PIN B AC GROUND

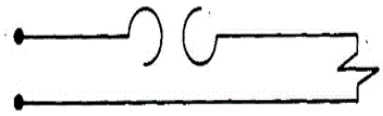


FIGURE 6. Electrical schematic for pin A and pin B.

7. The battery shall conform to military specification MIL-PRF-8565, except as modified by the following:

3.5.10.a Add "Heater current shall not exceed 6A at 115V." to the end of the paragraph.

3.8.7 Constant voltage discharge. Delete "in addition, the 60-second average current and total ampere-hour capacity requirements are not applicable." and substitute: "The discharge current shall not drop below the ampere values shown in figure 4 during the 14.0 volt constant voltage discharge. Inability of the sample to deliver the specified minimum

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discharge current for 1 minute shall constitute a test failure."

3.8.9 Cycling. In the first sentence, delete "100 cycles" and substitute "85 cycles".

3.8.13.1 Ground storage. Delete the text and add: "The battery shall be capable of storage at temperature conditions from -26ºC to 50ºC for up to 30 days and maintain a sufficient state of charge to successfully complete three start cycles per figure 3 herein."

3.8.16 Output performance. Delete the paragraph in its entirety and substitute the following:

3.8.16 Standby power output performance.

3.8.16.1 The battery, when discharged into a 16-ohm resistive load, shall maintain a terminal potential of 24.0 volts or greater for the discharge time and temperature conditions shown below, when tested in accordance with 4.5.28 of MIL-PRF-8565 as modified here.

TEMPERATURE (ºC)	DISCHARGE TIME (Hours)
-26	.5
-18	1.0
0	3.0
25	6.0
49	6.0

3.8.16.2 The battery, after completing a discharge into a 16-ohm load for the time and temperature conditions shown above, shall successfully complete three start cycles, with the energy required in figure 3, when tested in accordance with 4.5.28 of MIL-PRF-8565 as modified herein.

8. The battery shall conform to MIL-PRF-8565, except as modified by the following:

3.8.17 Cold and hot temperature charge. Delete the paragraph in its entirety and substitute the following:

3.8.17 Cold and hot temperature charging.

9. Add the following new paragraphs:

3.8.17.1 Half-hour charging. The battery, having 7.0 Ah capacity at temperatures in the range of -18 ºC to 55 ºC, shall be capable of delivering 14.0 Ah capacity after a 28.25 ± charging source has been applied to the battery for 30 minutes, when tested in accordance with 4.5.29, as modified herein.

3.8.17.2 One-hour charging. The battery, when discharged at temperatures in the

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range of -40 ºC to -19 ºC at 15.0A to 18.0V, shall be capable of delivering 14.0 Ah capacity after a 28.25 ±0.5V charging source has been applied to the battery for 1.0 hour. The battery shall not be damaged when a 28.25 ± 0.5V charging source is applied continuously to the battery at -40ºC, when tested in accordance with 4.5.29.2.1, as modified herein.

10. The battery shall conform to MIL-PRF-8565, except as modified by the following:

4.5.10 Cycling test. In the second sentence, delete "100 cycles" and substitute "85 cycles".

4.5.12 Charge and discharge test at low temperature. Delete paragraphs 4.5.12 and paragraphs 4.5.12.a through 4.5.12.g in their entirety.

4.5.19 Vibration test. Delete the paragraphs 4.5.19 and 4.5.19.a through 4.5.19.h in their entirety and substitute the following:

4.5.19 Endurance and performance random test. The battery assembly, including connectors and cable, shall be mounted to the test apparatus, per 4.4.5, and subjected to the random vibration test procedure for equipment installed in jet aircraft, Procedure I.A, Method 514.2 of MIL-STD-810. The battery shall be vibrated along each orthogonal axis in accordance with the Random Vibration Test levels specified in figure 5 of this specification sheet for a period of 21 minutes per axis. The battery heaters shall be energized and the battery charge at 28.25 ± 0.5 volts during vibration and shall be discharged at 97.5A for 1 minute prior to beginning each axis of vibration. The cables connected to the battery shall consist of four size AN-6 cables (two connected to the positive terminal and two connected to the negative terminal) and shall be clamped to the test apparatus at a cable length of 46 ± 3 centimeters from the battery."

4.5.22 Ground storage test. Line 4.5.22.d delete and add: "Subject the battery to three start cycles per figure 3, with a 2-minute rest period between starts.

4.5.28 Output performance test. Delete paragraphs 4.5.28, 4.5.28.a, through 4.5.28.d in their entirety and substitute the following:

4.5.28 Standby power output performance test.

4.5.28.1 Performance at -26 ºC. Battery sample 2 shall be charged per 4.4.2, and then placed on open-circuit at -26º for 12 ± .5 hours. The battery while remaining in the -26 ºC chamber, shall be discharged into a 16-ohm resistive load for 0.5 hour. The battery shall maintain a terminal potential of 24.0 volts or greater during the discharge. The battery shall be removed from the chamber and immediately subjected to three start cycles per figure 3, with a 2-minute rest period between starts. The battery shall meet the requirements of 3.8.16.

4.5.28.2 Performance at -18 ºC. Battery sample 3 shall be tested in accordance with 4.5.28.1, except that the test temperature shall be -18 ºC, and the discharge time into 16 ohms shall be 1.0 hour.

4.5.28.3 Performance at 0 ºC. Battery sample 1 shall be tested in accordance with

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4.5.28.1, except that the test temperature shall be 0 ºC, and the discharge time into 16 ohms shall be 3.0 hours.

4.5.28.4 Performance at 25 ºC. Battery sample 2 shall be tested in accordance with 4.5.28.1, except that the test temperature shall be 25 °C, and the discharge time into 16 ohms shall be 6.0 hours.

4.5.28.5 Performance at 49 °C. Battery sample 1 shall be tested in accordance with 4.5.28.1, except that the test temperature shall be 49 °C, and the discharge time into 16 ohms shall be 6.0 hours.

4.5.29 Cold and hot temperature charge test. Delete paragraphs 4.5.29, 4.6.29.a through 4.5.29.g in their entirety and substitute the following:

4.5.29 Cold and hot temperature charge test.

4.5.29.1 Half-hour charging. Battery sample 2 shall demonstrate the cold and hot temperature charging requirements of 3.8.17. The test shall be conducted as follows:

a. Charge battery at 28.25 ± 0.5V for 2 hours at 25 ºC.

b. Stabilize the battery at -18 ºC, then discharge at 15.0A to 18.0V. The battery shall provide at least 7.5 Ah capacity.

c. Stabilize the battery at 25 ºC, then charge at 28.25 ± 0.5V for 2 hours.

d. Stabilize the battery at -18 ºC, then discharge at 15.0A to remove all but 7.5 Ah, as determined by 4.5.29.1.b.

e. Rest battery to stabilize at test temperature, then apply 28.25 ± 0.5V for 30 minutes.

f. Immediately following charge, remove battery from test chamber and discharge at 15.0A to 18.0V. The battery must deliver 14.0 Ah minimum.

4.5.29.2 The battery shall be tested in accordance with 4.5.29.1 except that the test temperature shall be 55 ºC.

4.5.29.2.1 One hour charging. Battery sample 3 shall be subjected to a -40 ºC charge demonstration test and shall be conducted as follows:

a. Charge the battery at 28.25 ± 0.5V for 2 hours at 25 ºC.

b. Stabilize the battery at -40 ºC, and then discharge at 15.0A to 18.0V.

c. Rest battery to stabilize at -40 ºC, then apply 28.25 ± 0.5V for 1.0 hour.

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d. Immediately following charge, remove battery from test chamber and discharge at 15.0A to 18.0V. The battery shall deliver 14.0 Ah minimum.

Referenced documents: In addition to MIL-PRF-8565, this document references:

MIL-STD-810
MIL-PRF-18148/3

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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TABLE I. Inspection of batteries and order of tests.

Inspection number	Inspections	Qualification inspection			Requirements Paragraph	Method of inspection paragraph	Quality conformance Inspection		
		Qualification sample number					Group A tests	Group B tests	
		1	2	3				1	2
1	Visual and mechanical examination	X	X	X	3.5, 3.6, and 3.8.2	4.5.2	X	X	X
2	Dimensions and weight	X	X	X	3.8.3	4.5.3	X	X	
3	Dielectric strength	X	X	X	3.8.4	4.5.4	X	X	X
4	Strength of vent tubes	X	X	X	3.8.5	4.5.5	X	X	X
5	Color and marking	X	X	X	3.6	4.5.2	X	X	X
6	Electrical test preparation	X	X	X		4.5.6		X	X
7	Capacity discharge	X	X	X	Requirement 1 herein, 3.8.6, and requirement (1) of table III	4.5.7	X	X	X
8	Constant voltage discharge	X	X		3.8.7	4.5.8	X	X	
9	Strength of Receptacle			X	3.8.8	4.5.9			X
10	Standby power output performance	X	X	X	3.8.16	4.5.28			
11	Cold and hot temperature charge	X	X	X	3.8.17	4.5.29			
12	Cycling	X			3.8.9	4.5.10		X	
13	Capacity discharge at 49 °C		X		3.8.6 and requirement (2) of table III	4.5.11			
14	Discharge while inverted		X		3.5.9 and requirement (5) of table III	4.5.13			X
15	Altitude		X		3.7 and 3.5.9.	4.5.14			X
16	Mechanical shock	X			3.7 and 3.8.10	4.5.15			
17	Temperature shock	X			3.7 and 3.8.10	4.5.16			
18	Temperature rise and float		X		3.8.11	4.5.17			X
19	Battery gas emission		X		3.8.12	4.5.18			

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TABLE I. Inspection of batteries and order of tests - Continued.

Inspection number	Inspections	Qualification inspection			Requirements paragraph	Method of inspection paragraph	Quality conformance inspection		
		Qualification sample number					Group A tests	Group B tests	
		1	2	3				1	2
20	Vibration		X		3.7 and 3.8.10	4.5.19			X
21	Humidity		X		3.7 and 3.8.10	4.5.20			
22	Salt fog	X			3.7 and 3.8.10	4.5.21			
23	Ground storage			X	3.8.13.1	4.5.22			
24	Physical integrity at high temperature	X			3.8.7 and requirement (1) of table III	4.5.23			
25	Final examination	X	X	X	3.8.4	4.5.31		X	X

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CONCLUDING MATERIAL

Custodians:
Navy - AS
DLA - CC

Preparing activity:
Navy - AS

Project 6140-2021-029

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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